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ROLLNO: A-18

BATCH:-A1

ASSINGMENT:-9

Aim - Building a Simple Data Science Pipeline: Integrate all learned techniques in previous labs into a complete workflow.

Q1. What is a Data Science Pipeline?

A Data Science Pipeline is a step-by-step process used to convert raw data into useful results or predictions.

Stages: Data Collection – collecting data (CSV, database) Data Cleaning – removing missing or wrong data EDA (Exploration) – understanding patterns in data Feature Engineering – creating new useful columns Encoding & Scaling – converting data into proper format Model Training – applying machine learning model Evaluation – checking model performance

- Example: Predicting student marks based on study hours.

Q2. What is EDA?

EDA (Exploratory Data Analysis) is the process of analyzing data to understand it before applying machine learning.

Types: Univariate Analysis – study one column

Example: checking distribution of marks

Bivariate Analysis – study relationship between two columns

Example: study hours vs marks

- It helps in finding patterns, trends, and errors.

Q3. Label Encoding vs OneHot Encoding

Feature	Label Encoding	OneHot Encoding
Definition	Converts categories into numerical labels	Converts categories into binary columns
Example	Red = 0, Blue = 1, Green = 2	Red = [1,0,0], Blue = [0,1,0], Green = [0,0,1]
Number of Columns	1 column	Multiple columns
Order Issue	Creates order (can mislead model)	No order is created
Suitable For	Ordinal data (Low, Medium, High)	Nominal data (Color, City)
Model Impact	Can cause wrong interpretation	Gives better accuracy in many cases

Q4. Why is Feature Scaling Required?

Feature scaling is used to bring all features to a similar range so that no feature dominates others during model training.

In real datasets, features have different ranges. For example, age may range from 18–60, while salary may range from 10,000–1,00,000. Without scaling, machine learning models may give more importance to larger values like salary and ignore smaller ones like age.

- Feature scaling helps in:

Improving model accuracy

Increasing training speed

Helping algorithms converge faster

- It is mainly required in algorithms like:

Linear Regression

KNN

SVM

Types of Feature Scaling

1. Normalization (Min-Max Scaling):

Scales values between 0 and 1 Formula: $X' = (X - \min) / (\max - \min)$ Used when data has no extreme outliers

2. Standardization (Z-score Scaling):

Converts data to mean = 0 and standard deviation = 1 Formula: $X' = (X - \mu) / \sigma$ Used in most real-world datasets (handles outliers better) Difference Feature Normalization Standardization Range 0 to 1 No fixed range Outliers Affected Handles better Use Case Neural networks Most ML models

Q5. What is Feature Engineering?

Feature engineering is the process of creating new features from existing data to improve the performance of a machine learning model.

Raw data is often not enough for accurate predictions. By transforming or combining features, we can provide better input to the model.

Importance Improves model accuracy

Helps in better pattern detection

Reduces noise in data

Makes models more efficient

Types of Feature Engineering

1. Creating New Features

Example: Total Marks = Math + Science + English

2. Transforming Features

Example: Applying log transformation to reduce large values

3. Extracting Features

Example: From Date → Day, Month, Year

4. Combining Features

Example: Income per person = Total income / family members

```
import pandas as pd
import numpy as np

from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("/content/train (1).csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs)	female	38.0	1	0	PC 175

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null    int64
1   Survived        891 non-null    int64
2   Pclass         891 non-null    int64
3   Name            891 non-null    object
4   Sex             891 non-null    object
5   Age            714 non-null    float64
6   SibSp          891 non-null    int64
7   Parch          891 non-null    int64
8   Ticket         891 non-null    object
9   Fare           891 non-null    float64
10  Cabin          204 non-null    object
11  Embarked       889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

```
df.isnull().sum()
```

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

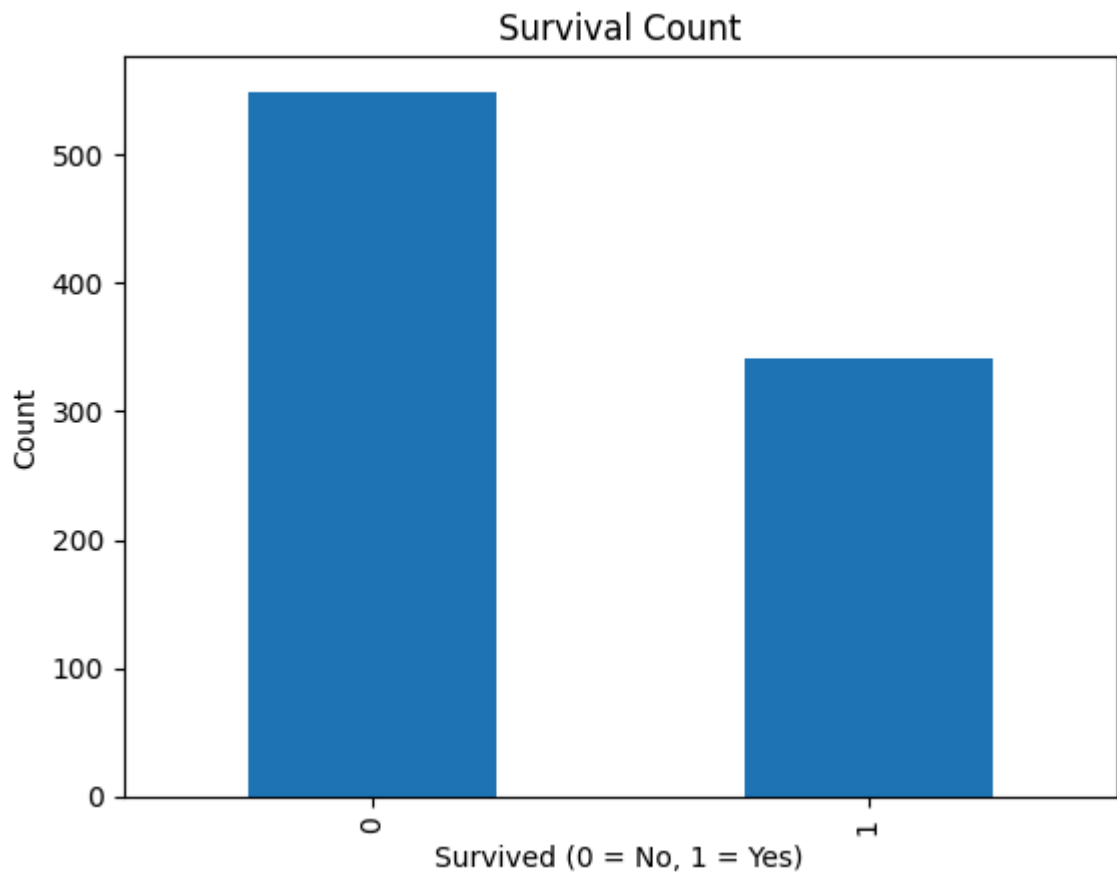
```
df.drop_duplicates(inplace=True)
```

```
df.corr(numeric_only=True)
```

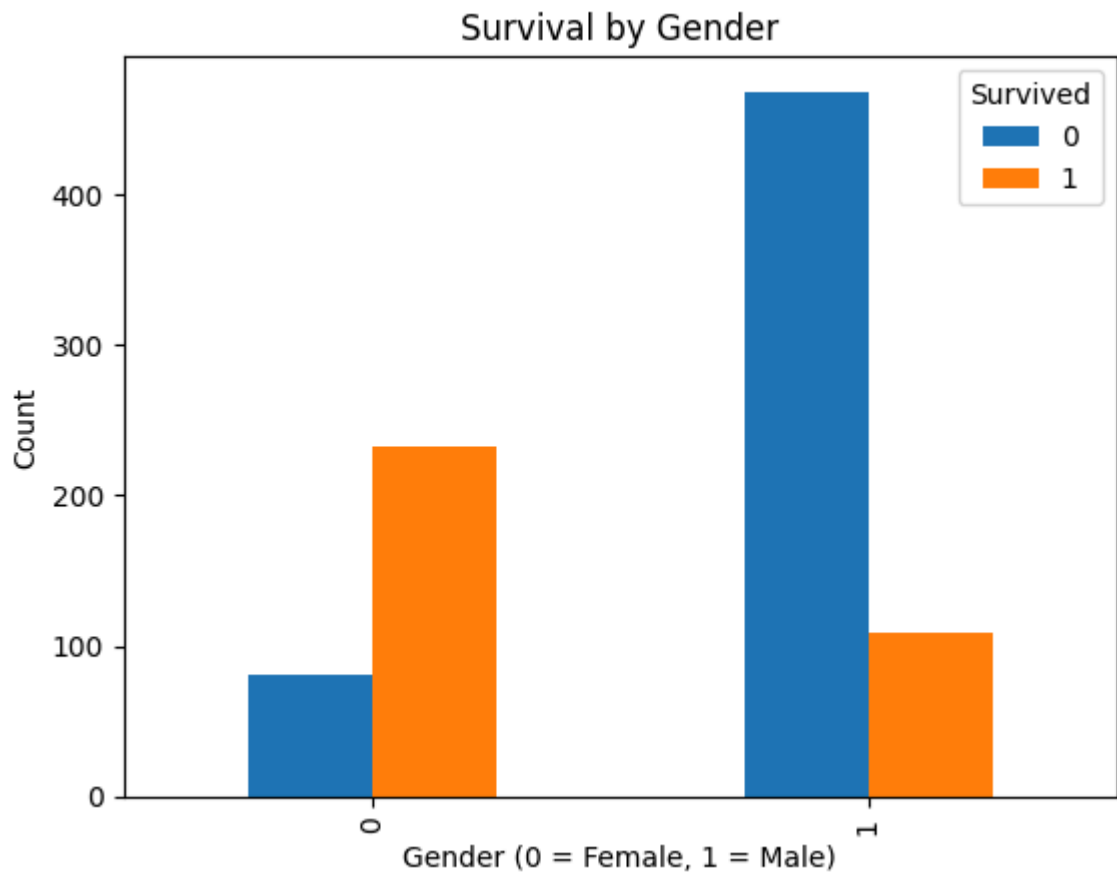
```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briaas	0	38.0	1	0	PC 17599

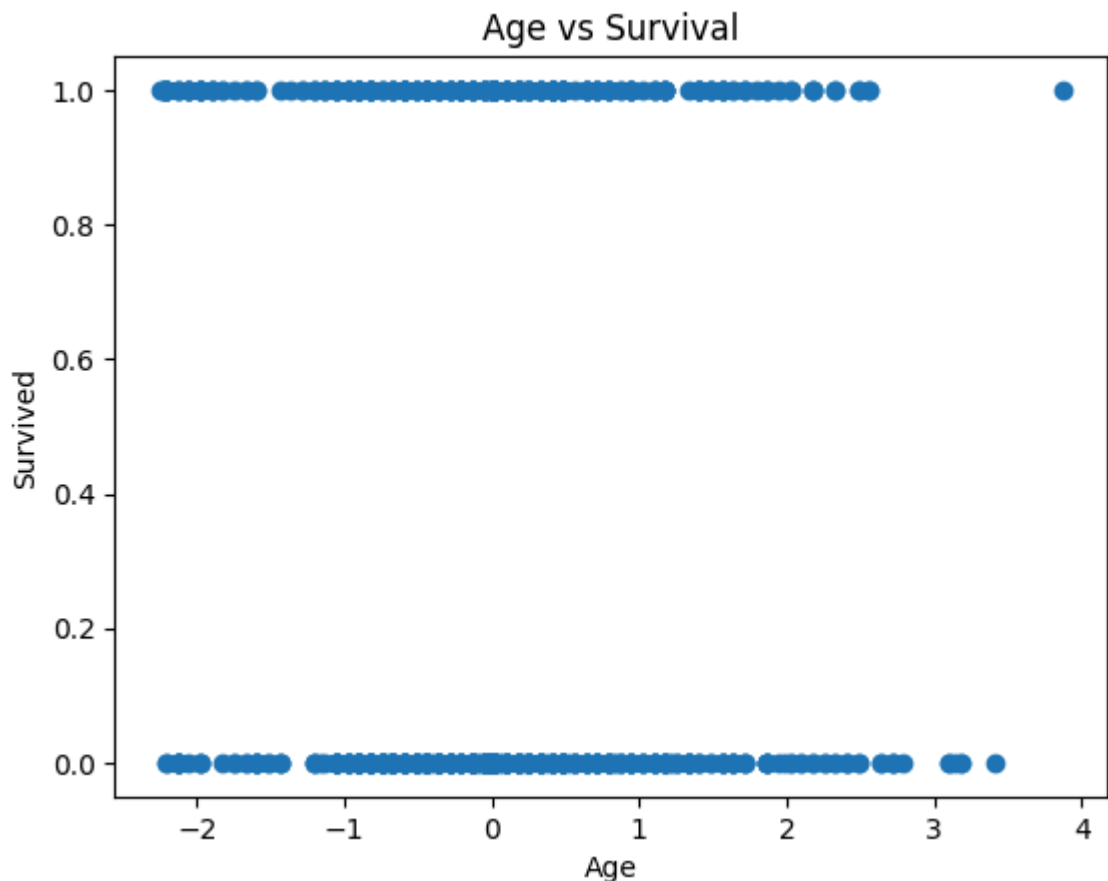
```
df['Survived'].value_counts().plot(kind='bar')
plt.title("Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()
```



```
pd.crosstab(df['Sex'], df['Survived']).plot(kind='bar')
plt.title("Survival by Gender")
plt.xlabel("Gender (0 = Female, 1 = Male)")
plt.ylabel("Count")
plt.show()
```



```
plt.scatter(df['Age'], df['Survived'])  
plt.title("Age vs Survival")  
plt.xlabel("Age")  
plt.ylabel("Survived")  
plt.show()
```



```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df['Sex'] = le.fit_transform(df['Sex'])
df['Embarked'] = le.fit_transform(df['Embarked'])
```

```
df['FamilySize'] = df['SibSp'] + df['Parch'] + 1
```

```
df['Age_Squared'] = df['Age'] ** 2
```

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

df[['Age', 'Fare', 'FamilySize', 'Age_Squared']] = scaler.fit_transform(
    df[['Age', 'Fare', 'FamilySize', 'Age_Squared']]
)
```

```
from sklearn.model_selection import train_test_split

X = df[['Pclass', 'Sex', 'Age', 'Fare', 'FamilySize']]
y = df['Survived']

X_train, X_test, y_train, y_test = train_test_split(
```

```
X, y, test_size=0.2, random_state=42
```

```
from sklearn.linear_model import LogisticRegression
```

```
model = LogisticRegression()
```

```
model.fit(X_train, y_train)
```

```
LogisticRegression ⓘ ?
```